

Chinmay Omkar Bitne

+1(623)-212-8463 | chinmaybitne82@gmail.com | www.linkedin.com/in/chinmaybitne/ | github.com/ChinmayBitne/ | chinmaybitne.github.io/

SUMMARY

Recent M.S. Information Technology graduate from Arizona State University focused on Generative AI, retrieval-augmented generation, and agentic LLM systems. Built RAG pipelines that cut document retrieval latency 65% and lifted semantic retrieval precision from 62% to 89% during an AI Engineering internship at Virtusa. Deployed a full-stack RAG application, AnswerLeaf AI, to production on Render and Vercel.

EDUCATION

Arizona State University

Masters, Information Technology

• GPA: 4.00

May 2026

Savitribai Phule Pune University

Bachelor's, Information Technology

• GPA: 3.64

Jun 2024

PROFESSIONAL EXPERIENCE

Integrated Travel

AI Engineer Intern

Jul 2026 - Present

Remote

- Developed and tested hybrid search and retrieval workflows using PostgreSQL/pgvector, implementing query normalization, input validation, ranking behavior, and filter composition for AI-powered search.
- Built an idempotent embedding pipeline with change-aware regeneration and traceability, preventing unnecessary embedding generation while maintaining consistency between source data and vector representations.
- Created comprehensive automated tests covering retrieval edge cases, model/database short-circuiting, filters, and ranking logic, while contributing text-cleaning and source-grounded summarization functionality to the analytics engine.

Virtusa Corporation

AI Engineer Intern

Apr 2024 - Jul 2024

Hyderabad, India

- Designed 2 RAG pipelines using LangChain and OpenAI GPT models for internal workflow use cases, reducing document retrieval latency by 65% and eliminating 40+ hours/month of manual lookups.
- Optimized transformer-based embedding models and prompt engineering pipelines, improving semantic retrieval precision from 62% to 89% for LLM-powered document automation workflows.
- Containerized two GenAI applications with Docker, set up CI/CD pipelines on Google Cloud Platform, and added evaluation-metric monitoring, which streamlined prototype delivery and ensured consistent performance during the internship.

PROJECT EXPERIENCE

AnswerLeaf AI - Grounded Academic Research Workspace

- Built and deployed a full-stack workspace (Render backend, Vercel frontend) for evidence-grounded assignments with source ingestion, hybrid retrieval, source allow lists, authority-aware reranking, and clickable citations.
- Implemented the orchestration and API layers with LangGraph and FastAPI, backed by PostgreSQL/pgvector for vector and keyword retrieval.
- Added citation verification and documented a scoped five-case synthetic regression test with retrieval hit rate and valid-citation checks.

ASTRA AI - Voice-First Windows Assistant

- Building a localhost-first Windows assistant with Gemini Live bidirectional audio in an Electron, React, TypeScript, Vite, and Express application.
- Connecting 50+ local tools for applications, files, games, media, system monitoring, reminders, and multi-step workflows.
- Integrating GCP for cloud-connected AI services while developing screen understanding, proactive alerts, Gmail/calendar briefings, and native system-tray integration.

NutriAssist AI - LLM-Powered Nutrition Assistant

- Fine-tuned Qwen with LoRA and prompt engineering, improving nutrition-recommendation accuracy from 71% to 92% on a domain-specific evaluation set.
- Designed an agentic multimodal pipeline orchestrating model interactions with external tools (Gemini API for meal image detection) and persistent session memory.
- Containerized the assistant with Docker and version-controlled deployment updates for local testing and reproducibility.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, TypeScript
- **AI / Machine Learning:** Large Language Models (LLM), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, OpenAI GPT, Agentic AI, Model Evaluation, Model Context Protocol (MCP), Embeddings, Prompt Engineering, Hugging Face, PyTorch, TensorFlow, Fine-tuning (LoRA)
- **Cloud & MLOps:** AWS, Google Cloud Platform (GCP), Docker, CI/CD, Git
- **Backend & Data:** FastAPI, REST APIs, Flask, PostgreSQL, pgvector, Vector Databases

ACHIEVEMENTS

- **IEEE Paper Publication:** Author, "Generative AI Solution for Lane Departure, Pedestrian Detection, and Paving of Autonomous Vehicle" ICRTC 2023.
- **Winner, Virtusa Jatayu S3 Hackathon:** Led ML pipeline development for predictive analytics using 5G sensor data.
- **Best Project Award, Ural Federal University (URFU):** Alternovation, an AI-powered furniture visualization and cost-estimation prototype.

CERTIFICATIONS

- AWS Academy Graduate: Machine Learning Foundations, Machine Learning for Natural Language Processing, Data Engineering
- Google: IT Automation with Python Professional Certificate